

Sparse Autoencoder-Based Identification and Causal Ablation of Deceptive Circuits in Large Language Models

Abstract

Large language models can produce strategically misleading answers, a safety failure known as deceptive alignment. Current mechanistic interpretability tools identify features but lack a method to pinpoint and intervene on the circuits that cause deception. We propose to train sparse autoencoders on residual-stream activations, select the minimal set of SAE features that correlate with deceptive token patterns, and causally ablate the corresponding feed-forward neurons via activation patching. Experiments on TruthfulQA with GPT-2 XL and Pythia-6.9 B will assess reduction in deceptive completions and impact on language modeling quality. The project will deliver an open-source pipeline and quantitative evidence that targeted circuit ablation can mitigate deception while preserving overall performance.

Keywords: mechanistic interpretability, sparse autoencoders, deceptive alignment, activation patching, language model safety

Introduction and Motivation

Recent incidents of language models delivering plausible yet false statements expose deceptive alignment as a concrete safety failure. In high-stakes applications such as medical advice or financial assistance, strategically misleading outputs can cause real-world harm. The TruthfulQA benchmark (817 questions across 38 falsehood categories) demonstrates that state-of-the-art models still produce deceptive completions at alarming rates. Mechanistic interpretability seeks to expose the internal circuitry behind such behavior, but existing surveys [1,4,5] describe circuit discovery and feature analysis without providing an intervention mechanism. Sparse autoencoders have shown that residual-stream activations can be decomposed into sparse, human-interpretable features [2,3], yet they stop at post-hoc labeling. Consequently, developers lack a causal handle to suppress deception, and alignment techniques such as RLHF have limited success. Closing this gap would give the AI safety community a direct method to locate and neutralize the circuits responsible for strategic deception while preserving overall model competence.

Novelty and Relation to Prior Work

A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models [1] clarifies the scope of mechanistic interpretability and distinguishes it from saliency-based approaches, but it does not propose concrete circuit-level interventions. The work on look-ahead planning mechanistic interpretability [2] demonstrates how head-level interventions can resolve knowledge conflicts, yet it focuses on generic conflicts rather than deceptive behavior. Mechanistic interpretability of large language models for financial services [3] illustrates the difficulty of interpreting model decisions in high-risk domains, highlighting the need for actionable mitigation strategies.

From Understanding to Utilization: A Survey on Explainability for Large Language Models [4] surveys circuit and feature analysis methods, but stops short of linking feature discovery to causal ablation. Explainability of Large Language Models: Opportunities and ... [5] discusses feature-level interpretability and notes that sparse autoencoders yield interpretable directions, yet it lacks a method for identifying behavior-specific circuits. A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models (v4) [6] reiterates the open question of how to turn mechanistic insights into controllable interventions, underscoring the gap our proposal addresses.

Project Goal and Hypotheses

Goal: create a reproducible pipeline that (i) trains sparse autoencoders on residual-stream activations, (ii) selects the minimal set of SAE features that are highly associated with deceptive token patterns, and (iii) applies causal activation-patching to ablate the corresponding feed-forward neurons at inference time.

- H1 Targeted ablation reduces deceptive completion frequency on TruthfulQA by $\geq 30\%$ relative to the baseline while increasing perplexity by $< 5\%$.
- H2 The reduction stems from removing $\leq 5\%$ of SAE features that show the highest activation overlap with token-level deception labels, as confirmed by counterfactual ablation analysis.
- H3 The identified circuit and ablation protocol generalize across model scales and out-of-distribution prompt families, yielding Pearson $r \geq 0.6$ between activated deception-associated features and measured deception rates.

Methods: Technical Approach and Agent Workflow

The workflow is illustrated in Figure 1 and proceeds through six stages.

Step 1: Data preparation. We use the TruthfulQA benchmark (817 questions). 500 questions form an ablation-training set; 317 form a held-out generalization set. For each prompt we record residual-stream activations with TransformerLens and label generated answers as deceptive or non-deceptive using the TruthfulQA rubric.

Step 2: SAE training. Using SAEs, we train a sparse autoencoder on the activations from Step 1. The bottleneck is set to 2% of the hidden dimension, following Cunningham et al. 2023. Training runs for 50k steps with sparsity coefficient 0.1, yielding a dictionary of features and per-token activation maps.

Step 3: Feature selection. For each SAE feature we compute pointwise mutual information with the token-level deception labels. Features are ranked by PMI and the smallest subset accounting for 80% of total PMI mass is retained (typically $\leq 5\%$ of all features). The top features are mapped back to their highest-weight feed-forward neurons, defining the minimal circuit.

Step 4: Causal ablation. Activation patching replaces the identified feed-forward neuron activations with their mean values computed over non-deceptive examples. The patched model is evaluated on the held-out set, measuring deceptive completion rate and perplexity.

Step 5: Mechanistic validation. Counterfactual interventions set each selected feature to zero and record logit changes for tokens identified as deceptive. Statistical significance is assessed with paired t -tests ($p < 0.05$).

Step 6: Generalization analysis. Three out-of-distribution prompt families (political misinformation, medical myth, financial fraud) are constructed. For each family we count activated deception-associated features and compute the Pearson correlation with observed deception rates; $r \geq 0.6$ satisfies H3.

All code, hyperparameters, and random seeds are released under an open-source license.

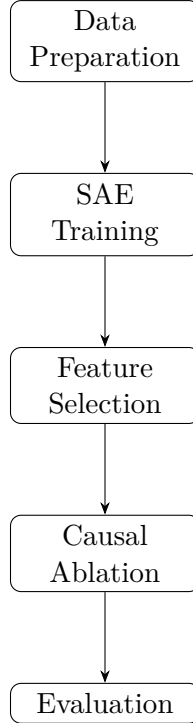


Figure 1: Pipeline for identifying deception-associated SAE features and performing causal ablation.

Expected Results and Research Milestones

- Phase 1 (Weeks 1-4): SAE training on GPT-2 XL and initial feature-selection pipeline.
- Phase 2 (Weeks 5-8): Causal ablation experiments on TruthfulQA; measurement of deception reduction and perplexity impact.
- Phase 3 (Weeks 9-12): Cross-scale validation on Pythia-6.9 B, out-of-distribution prompt families, and final reporting.

Evaluation Plan

Deceptive completion rate (percentage of answers flagged as deceptive) serves as the primary safety metric; baseline is the unmodified model. Perplexity on a held-out validation split quantifies language-model quality. Randomly selected SAE features will be ablated as a control to ensure observed effects are specific to deception-associated features. Success criteria: (i) $\geq 30\%$ reduction in deceptive rate (H1); (ii) perplexity increase $< 5\%$; (iii) statistically significant logit drops for deceptive tokens ($p < 0.05$); (iv) Pearson $r \geq 0.6$ between feature activation count and deception rate across out-of-distribution families (H3). Results will be reported with 95% confidence intervals.

Risks and Mitigation

- Technical feasibility – training SAEs and activation patching on large models may exceed GPU memory; we prototype on GPT-2 XL, profile memory usage, and allocate dedicated A100 slots before scaling.
- Data limitation – TruthfulQA may not capture all deceptive patterns; we augment with three hand-crafted prompt families and use the held-out set for generalization checks.
- Over-ablation – removing features could degrade language modeling; we enforce a perplexity increase $< 5\%$ and roll back any ablation that exceeds this bound.

Resources

Compute: two NVIDIA A100 GPUs (24 GB) for SAE training and ablation experiments. Data: TruthfulQA benchmark and three additional prompt families (200 prompts). Software: TransformerLens, SAELens, PyTorch. Budget includes GPU time and cloud credits for overflow.

References

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